

MOHAMMAD NAVED

Lead Systems & AI Engineer | Full-Stack Architect & Game Developer (UE5)

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PROFILE SUMMARY

Lead Systems & AI Engineer with a proven track record of shipping production-grade full-stack applications, real-time 3D interactive mechanics, and commercial IoT infrastructure. Experienced in Unreal Engine 5, Blender asset pipelines, and integrating Model Context Protocol (MCP) plugins to connect LLMs with game engines. Founded Codtrex AI, delivering autonomous AI agents and cloud systems, and achieved official Google Smart Home certification for 4Layers (ESP32 / AWS IoT Core). Deep proficiency spanning modern web (React 19, Next.js 15/16, TypeScript), backend microservices (FastAPI, PostgreSQL, Docker), and game/systems engineering (Unreal Engine 5, C++, Python).

CORE EXPERTISE

- Game Development & 3D Systems** - Unreal Engine 5 (Blueprints, Physics, Gameplay Mechanics), Blender (3D modeling, UVs), Model Context Protocol (MCP) tooling, C++
- AI & Autonomous Systems** - production Gemini API integration, multi-LLM orchestration, agentic workflows, MCP server design, prompt architecture
- Full-Stack Product Architecture** - React 19, Next.js 15/16, TypeScript, Tailwind CSS v4, Vite, Supabase (PostgreSQL), FastAPI (Python)
- IoT & Embedded Systems** - ESP32 firmware (Arduino/C++), MQTT protocol, AWS IoT Core, mTLS device authentication, OTA firmware updates
- Cloud & DevOps** - AWS (IoT Core, App Runner, ECR, S3), Docker, Vercel, GitHub Actions CI/CD pipelines
- Systems Ownership** - end-to-end engineering from requirements to deployment; Google Smart Home certification lifecycle

EXPERIENCE

Codtrex AI - Boutique AI Engineering Studio2025 - Present

Lead Engineer and Founder | codtrex.vercel.app | React 19, Vite, Tailwind CSS v4, Supabase, Gemini API, Vercel

- Founded and operate a boutique AI engineering studio offering end-to-end LLM integration, full-stack web products, IoT systems, and technical consultation to clients
- Built and shipped the studio public website with an AI chat assistant, service tiers, and client case studies deployed at codtrex.vercel.app
- Designed multi-LLM product architectures and delivered client-facing AI solutions across web, mobile, and embedded platforms

4Layers - Google Certified Smart Home IoT Ecosystem2025 - Present

Lead Systems Engineer and Founder | ESP32, AWS IoT Core, MQTT, mTLS, React Native, FastAPI, Docker, AWS App Runner

- Achieved official Google Smart Home certification - passed 71 out of 75 tests in the Google Home Test Suite; devices listed and functional in the Google Home ecosystem
- Architected 3-source real-time state synchronization (ESP32 hardware to AWS IoT Core MQTT to Google Home) achieving zero state divergence across physical buttons, app controls, and voice commands
- Engineered QA Hold Mode to prevent Google Home from reverting local hardware states during certification test runs - a critical blocking issue independently discovered and resolved
- Built OTA 2.0 with version pinning and rollback safety for wireless firmware updates across all deployed devices without physical access
- Designed custom ESP32 hardware: 4-channel relay switch boards and Triac-based multi-speed fan regulator; mTLS certificates provisioned per device via AWS IoT Core

MediStock - Cloud Pharmacy Management SaaS2025 - Present

Lead Engineer and Founder | React 18, Supabase, PostgreSQL, Capacitor Android, Vercel | medistock-pharma.vercel.app

- Built and shipped a full B2B pharmacy SaaS covering inventory management, expiry tracking, billing, and supplier management for pharmacy businesses
- Deployed cross-platform as a live web application and Android APK via Capacitor; targeting Google Play Store submission
- Designed a multi-tenant Supabase (PostgreSQL) schema with Row Level Security ensuring complete pharmacy data isolation across accounts

PROJECTS

UsedTech Market - C2C Electronics Marketplace2025

Product Architect | Next.js 15/16, React 19, TypeScript, Supabase, Tailwind CSS v4

- Architected a consumer-to-consumer marketplace for buying and selling used electronics with real-time listings, advanced search filters, and seller dashboards
- Built on Next.js 15/16 App Router with server-side rendering for SEO-optimized product pages and Supabase real-time subscriptions for live listing updates
- Designed the full product data model, authentication flow, and listing lifecycle from draft to published to sold states

AI-Assisted 3D & Gameplay Systems2025 - Present

Systems & Gameplay Developer | Unreal Engine 5, Blender, Model Context Protocol (MCP), Python, C++

- Built real-time interactive gameplay mechanics, character locomotion physics, and environment blockouts in Unreal Engine 5 using custom Blender 3D assets
- Integrated Model Context Protocol (MCP) plugins to connect LLMs with Unreal Engine runtime, enabling AI-assisted world generation and agentic NPC behavior pipelines
- Architected modular Blueprint and C++ components for event-driven gameplay logic and rapid procedural prototyping

DataLens AI - Full-Stack AI Analytics Platform2025

Self-Initiated | React, TypeScript, Recharts, Gemini AI, DuckDB, SQL

- Built an AI analytics platform with a natural-language-to-SQL chat interface enabling non-technical users to query data through plain English questions
- Developed ML Lab module for model output explanation and business insight surfacing using Gemini-powered prompt templates
- Designed the ETL pipeline, SQL engine, and data visualization layer end-to-end as a single engineer

Interactive Portfolio and Automation Pipeline2025

Self-Initiated | React 19, TypeScript, Tailwind CSS v4, Node.js, Puppeteer, Gemini AI

- Built a single-page interactive portfolio with hardware-accelerated animations, an AI chatbot powered by Gemini, and a context-aware navigation system
- Automated resume PDF generation via a Node.js and Puppeteer script producing a pixel-perfect A4 PDF from live HTML source

TECHNICAL SKILLS

Game Dev & 3D: Unreal Engine 5, Blender (3D Modeling), Model Context Protocol (MCP), Gameplay Physics, Blueprints

AI & Autonomous: Gemini API, Multi-LLM Orchestration, Prompt Engineering, Agentic Workflows, MCP Servers

Frontend: React 19, Next.js 15/16, TypeScript, Tailwind CSS v4, Vite, Framer Motion, Recharts

Backend: FastAPI (Python), Node.js, REST APIs, Supabase, PostgreSQL, SQL, DuckDB

IoT & Embedded: ESP32 (Arduino/C++), MQTT, AWS IoT Core, mTLS, OTA Firmware Updates

Cloud & DevOps: AWS (IoT Core, App Runner, ECR, S3), Docker, Vercel, GitHub Actions CI/CD

Languages: TypeScript, Python, C++, SQL, JavaScript

EDUCATION

B.Tech, Computer Science - RGPV University, Indore2025

Rajiv Gandhi Proudyogiki Vishwavidyalaya | Madhya Pradesh, India

CERTIFICATIONS AND ACHIEVEMENTS

- Google Smart Home Certification - 4Layers IoT Ecosystem (71 out of 75 tests passed in the Google Home Test Suite)
- Generative AI with Gemini - Prompt Engineering and LLM Application Development
- Python for Data Analysis and Visualization - Self-directed
- SQL for Data Analytics - Relational Database Workflows
- Power BI - Interactive Dashboard Design and Reporting